

At least one independent sharp leg in each pair; full N3 responses

detector 1

$$z_{D1} = \|b_{D1}\|/\|r_{D1}\|$$
$$0 < z_{D1} \leq 1$$

detector 2

$$z_{D2} = \|b_{D2}\|/\|r_{D2}\|$$
$$0 < z_{D2} \leq 1$$

source 1

$$z_{S1} = \|b_{S1}\|/\|r_{S1}\|$$
$$0 < z_{S1} \leq 1$$

source 2

$$z_{S2} = \|b_{S2}\|/\|r_{S2}\|$$
$$0 < z_{S2} \leq 1$$

Unit Born readouts fix overlaps; all four legs combine

$$p_j = z_j^2, \quad z_{\text{sharp}} = 1, \quad Z = z_{D1} z_{D2} z_{S1} z_{S2}$$

The actual coefficient and norm difference return exactly

$$C_{\text{full}} = Z C_{\text{base}}$$
$$\|C_{\text{full}} - C_{\text{base}}\| = (1 - Z) \|C_{\text{base}}\|$$